SASTRA DEEMED TO BE UNIVERSITY
(U/S 3 of the UGC Act, 1956)
THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

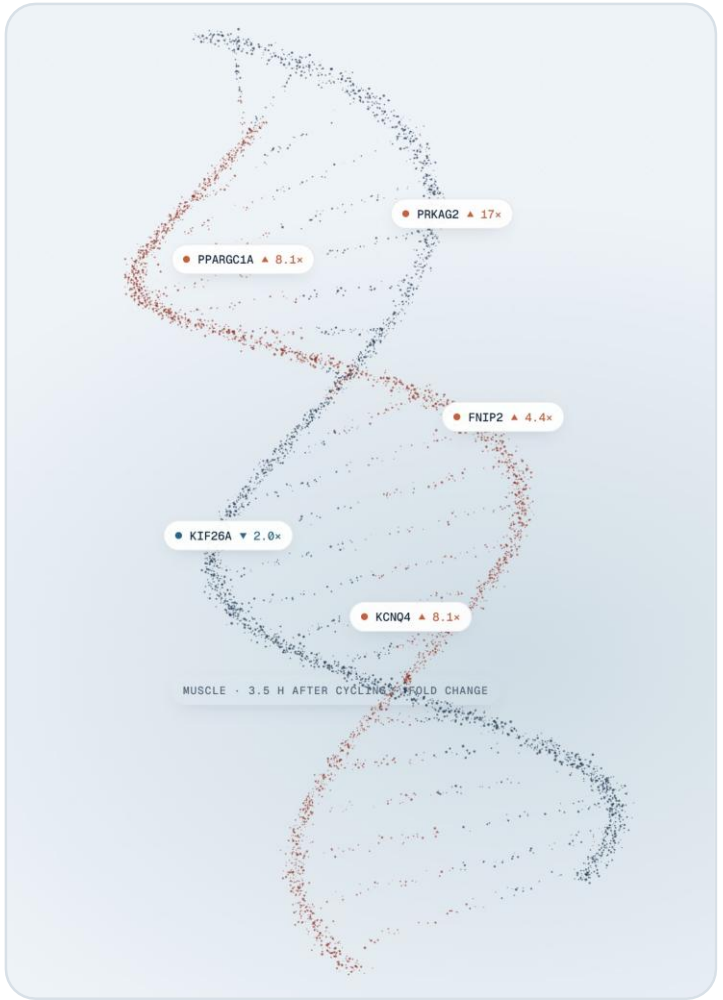
temenos

TEJAS 2026 GRAND FINALE · AI FOR DRUG DISCOVERY & PERSONALIZED MEDICINE

SynThera

Finding approved drugs that copy exercise

A human-data, AI-powered screen of 29,680 compounds



Team TEJAS019 · SASTRA Deemed University, B.Tech Bioinformatics

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Exercise is medicine, but millions cannot take it

Exercise lowers the risk of almost every chronic disease. For many patients it is impossible.

1.8 billion

adults worldwide (31%) are not active enough. In South Asia it is 45%.

WHO, 2024 release (2022 data)

Patients who cannot exercise at all

Spinal cord injury	Advanced heart failure
Intensive care & surgery	Cancer cachexia
Frail elderly	Long bed rest

THE GAP

Every published computational exercise-mimetic screen starts from rat or mouse data and assumes it transfers to humans. That assumption was never tested, because comparable human data did not exist.

OUR QUESTION

Which approved drugs switch genes on and off the way a workout does, in real humans?

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SynThera: from a human workout to a drug shortlist

One pipeline, five stages, built entirely on open data.

STEP 1

Exercise fingerprints

MoTrPAC: 175 adults, endurance and resistance, 3 tissues, 3 time points. 18 fingerprints.

STEP 2

Drug fingerprints

LINCS L1000: 720,216 lab profiles, one per compound. 29,680 compounds.

STEP 3

AI scoring

4 methods incl. the GSFM foundation model, merged by Robust Rank Aggregation.

STEP 4

Benefit filter

Keep exercise's adaptations, reject its stress. Cell-killing drugs removed.

STEP 5

Shortlist

Approved drugs for lab testing: 82 for endurance, 77 for resistance (weight) training.

Outputs ranked atlas of 29,680 compounds × 18 fingerprints · interactive website · FAIR open data (JSON-LD) · rat comparison arm

Six open datasets, one reproducible 10-step pipeline

Heavy computation runs once; every result is saved, versioned and traceable.

175
adults, vs a rested group

18
exercise fingerprints

720,216
drug profiles read (33 GB)

29,680
compounds scored

1,595
approved drugs among them

~430 M
scores computed

Dataset	What it gives us
MoTrPAC human (c2.0)	**Exercise fingerprints**
CMap LINCS 2020 L1000	**Drug fingerprints**
Drug Repurposing Hub	**Approval status, mechanism**
MSigDB Hallmark + Reactome	**Adaptive vs stress genes**
MoTrPAC rat (PASS1B)	**Comparison arm**
GSFM, Hugging Face	**Pretrained AI model**

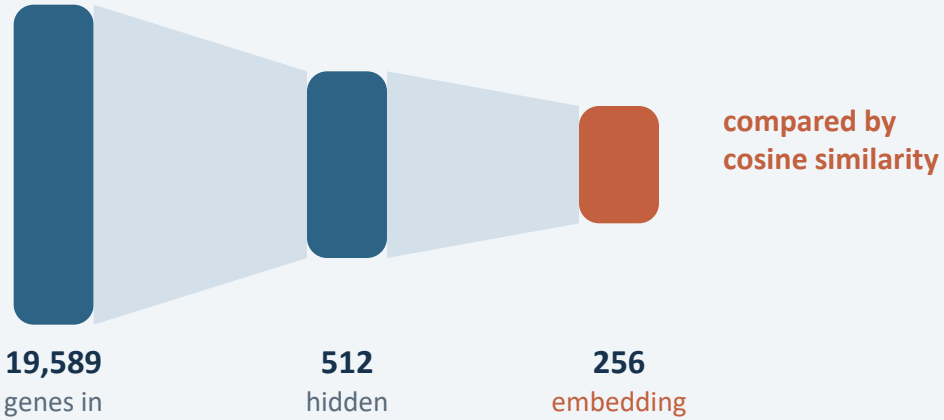
Stack Python 3.12 · NumPy · SciPy · pandas · PyTorch · UMAP · HTML/JS

Speed 29 min on a 6-core laptop; seed 19 gives identical runs

AI reads gene lists by meaning, and four methods vote

GSFM · Gene Set Foundation Model

Ma'ayan Lab, Patterns 2026 · autoencoder · 15.3 M parameters



Turns any gene list into 256 numbers. Lists about the same biology land close together, even with few genes in common. Used as published: transfer learning, no retraining.

Four scorers, one fair vote

Classic CMap
WTCS

Rank correlation
Spearman

Extreme genes
XSum

AI model
GSFM

Fair scoring each drug vs its own 200 random fingerprints

Robust Rank Aggregation merges the 4 ranks, then FDR

−38% top-50 overlap without GSFM: biggest drop

Three design choices that set SynThera apart

Each one changes which drugs reach the shortlist.

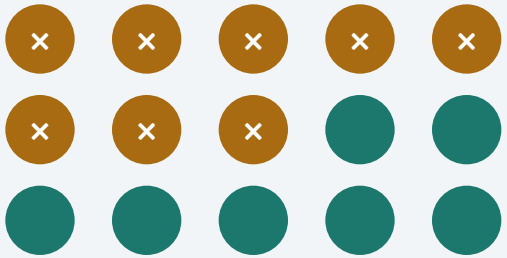
Human data first



0 drugs in common

Same pipeline, rat data: a completely different list (r = −0.20). 36% of the rat signal is lost converting genes to human.

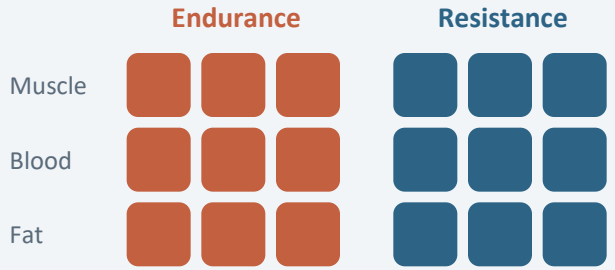
Benefit, not stress



8 of 15 top hits are cell-killing drugs

Benefit index: 701 adaptive vs 872 stress genes. Drugs that copy only the stress of exercise are filtered out.

Two exercise types



18 fingerprints: 2 × 3 tissues × 3 times

Two shortlists: endurance and resistance training are screened separately, so muscle-building gets its own list.

82 and 77 approved drugs worth testing in the lab

Top candidates by benefit index. All are already approved for other uses.

Endurance (cycling)

#	Drug	Used for	Benefit
1	**Sitagliptin**	Diabetes	**+0.78**
2	**Tioconazole**	Yeast infection	**+0.62**
3	**Olmesartan**	Hypertension	**+0.57**
4	**Phenytoin**	Seizures	**+0.55**
5	**Mepacrine**	Giardiasis	**+0.50**
6	**Sumatriptan**	Migraine	**+0.49**

Resistance (weight training)

#	Drug	Used for	Benefit
1	**Phenytoin**	Seizures	**+1.11**
2	**Sitagliptin**	Diabetes	**+1.07**
3	**Sertraline**	Depression	**+0.87**
4	**Thiothixene**	Schizophrenia	**+0.86**
5	**Protriptyline**	Depression	**+0.76**
6	**Sumatriptan**	Migraine	**+0.75**

105 different approved drugs

54 on both lists

Top 5% of 29,680 to qualify

Leads for lab testing, not treatments.

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Can we trust the results? Four simple tests

We tested SynThera on questions where we already know the right answer.

Does it spot the opposite of exercise? PASSED

Weight training builds muscle through mTOR. All 4 mTOR-blocking drugs ranked in the top 0.4% most opposite to it, unprompted.

Does it hold up when data is messy? PASSED

With 20% of the data scrambled, the top 50 still overlaps 48–77% with the original list. By chance: about 2%.

Do the four methods agree? PASSED

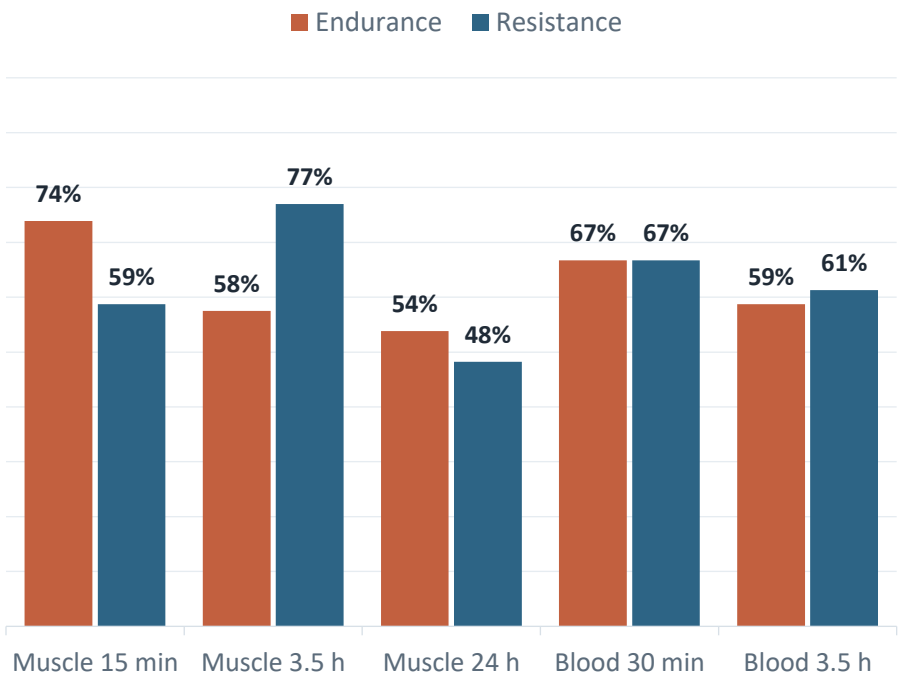
Mostly (r = 0.46–0.65): the same overall picture, yet each method adds something of its own.

Does it find known 'exercise pills'? PARTLY

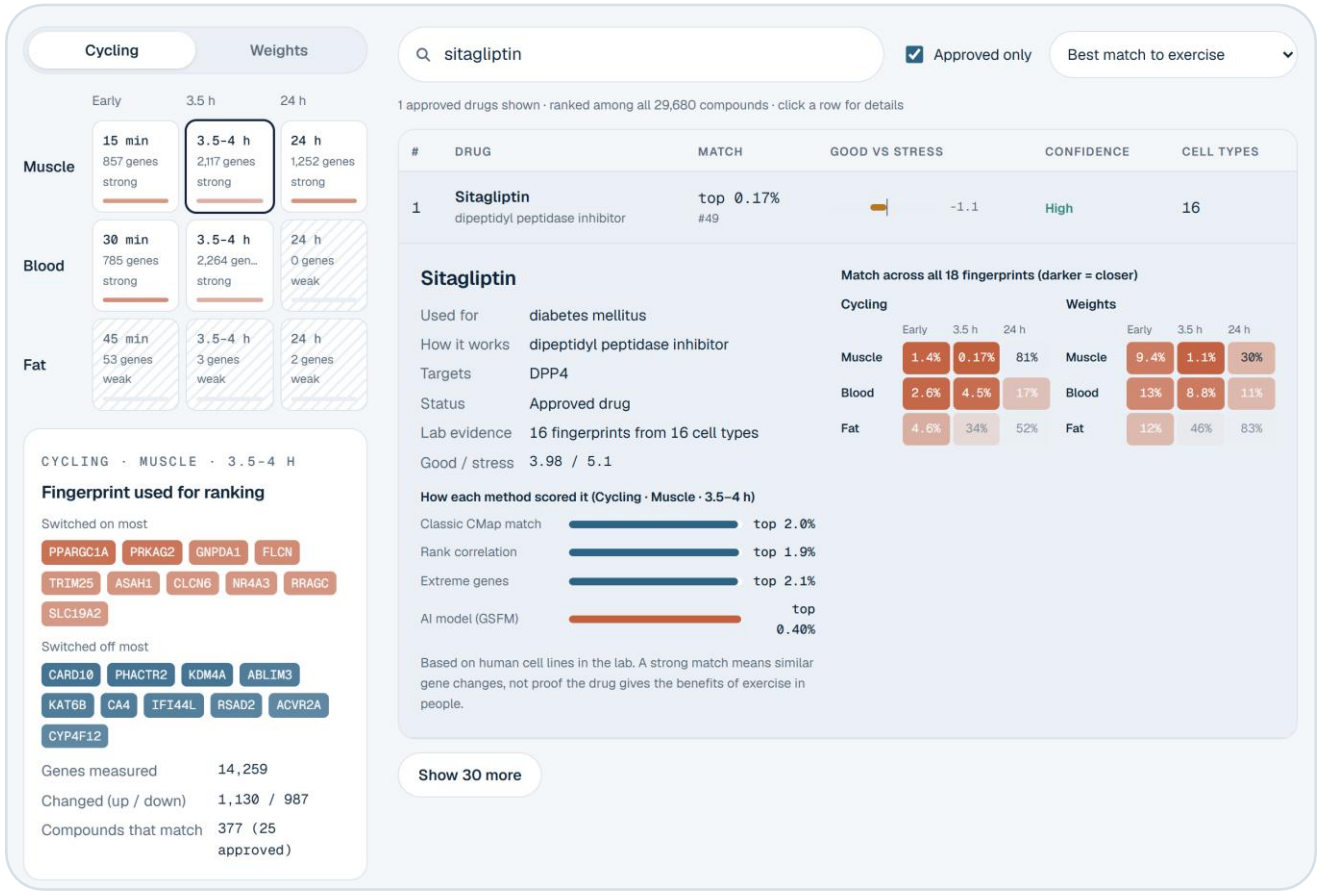
β2-agonists return for weight training; GW-501516 reaches the top 9–12% in muscle; metformin and AICAR do not.

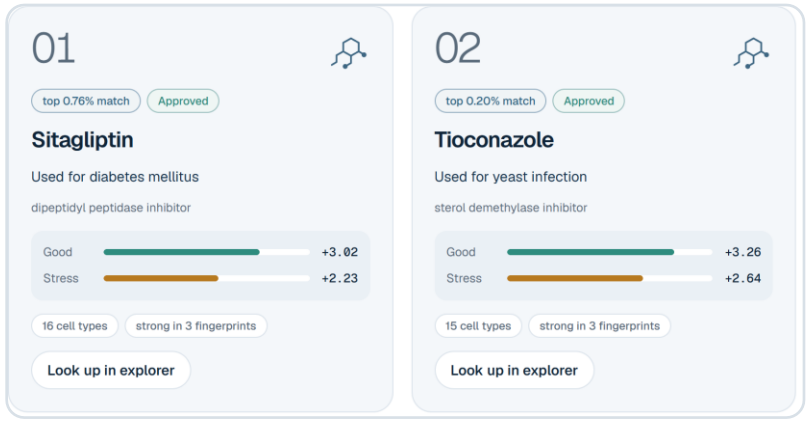
Test 2 in detail: the list holds steady

Top-50 overlap with the original · 20% of data scrambled, 30 reruns

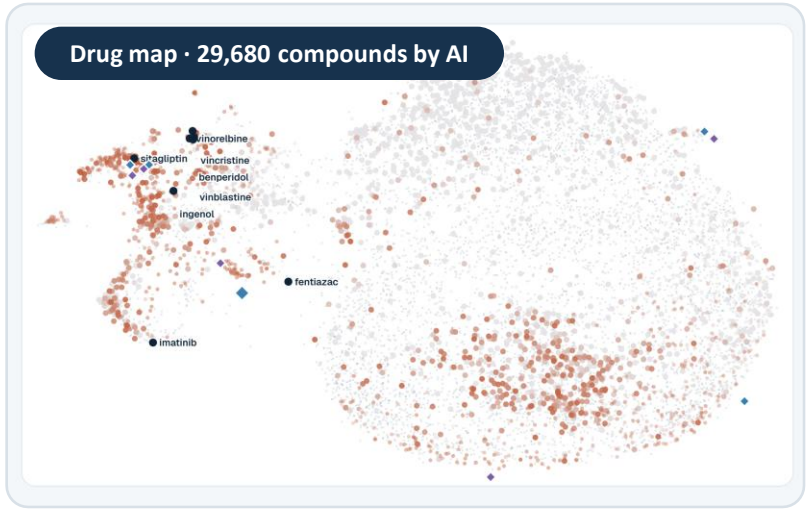


Working prototype: every result, live and explorable





Shortlist each candidate's good vs stress balance



LIVE DEMO **Explorer** sitagliptin in all 18 fingerprints · localhost:8000, works offline

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Impact today, and how SynThera scales

Impact

- Patients who cannot exercise get candidates already approved, with known safety records
- Researchers get an open FAIR atlas of 29,680 compounds
- A plain-language website anyone can explore

Scalability

- Any human gene signature plugs in: new tissues, diseases, long-term training data
- Each fingerprint scores independently; 29 min on a laptop today
- Static website: free to host anywhere

Next steps

- Test top candidates in human muscle cells
- Add protein and metabolite layers
- Partner with clinicians for patient groups

SDG 9 **Industry, Innovation & Infrastructure** Sub-theme: AI for Drug Discovery & Personalized Medicine

Honest limits one exercise session, not months of training · drug data from cancer cell lines · predictions to test, not treatments

Thank you. Questions welcome.